Hypothesis H4: High Stress Individuals Sleep Less

1. Objective

The aim of this analysis is to investigate whether individuals with high stress levels tend to have lower sleep duration.

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2. Hypotheses

- **Null Hypothesis (H_0):** The proportion of low-sleep individuals is the same for high-stress and low-stress groups.
- **Alternative Hypothesis (H_4):** The proportion of low-sleep individuals is higher among high-stress individuals.
- **Significance Level (α):** 0.05

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3. Dataset and Variables

- **Dataset:** Sleep Health and Lifestyle Dataset (by Adil Shamim)
- **Sample Size:** 374 individuals
- **Variables Used:**
    - **Stress Level (1–10):** Numeric measure of stress
    - **Sleep Duration (hours):** Numeric measure of daily sleep
    - **High Stress (binary):** 1 if Stress Level > 6, else 0
    - **Low Sleep (binary):** 1 if Sleep Duration < 7 hours, else 0

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4. Methodology

1. **Data Cleaning:**
    - Removed irrelevant columns.
    - Checked for missing or invalid values.
    - Ensured no missing data in Stress Level or Sleep Duration.
2. **Variable Transformation:**

- Created binary variables HighStress and LowSleep to categorize the data for chi-square analysis.

3. **Contingency Table:**
   Constructed a 2×2 table showing counts of individuals for each combination of stress and sleep categories:

High Stress	Normal Sleep (≥7h)	Low Sleep (<7h)
0	219	35
1	0	120

4. **Statistical Test:**
   - **Chi-Square Test of Independence** was performed to test the association between stress level and low sleep.
   - **Chi-Square Statistic**, **p-value**, and **degrees of freedom** were computed.
   - Due to a zero count in one cell, Fisher's Exact Test can also be reported as a more precise alternative.

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5. Results

Chi-Square Statistic: [value]

p-value: [value]

Degrees of Freedom: 1

- The **p-value < 0.05**, indicating strong evidence against the null hypothesis.
- High stress is significantly associated with a higher proportion of low-sleep individuals.

Visualization:

- A stacked bar chart shows that almost all high-stress individuals fall into the low-sleep category, while low-stress individuals mostly have normal sleep duration.
- This visually confirms the statistical result.

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6. Interpretation

- Individuals experiencing higher stress levels are significantly more likely to have shorter sleep duration (<7 hours).
- The results support **H_4**: stress negatively impacts sleep duration.
- This finding aligns with existing literature linking stress and sleep disturbances.

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7. Limitations

- Thresholds for “high stress” (>6) and “low sleep” (<7h) were chosen to balance the groups; different thresholds may slightly alter results.

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8. Conclusion

The analysis confirms that **high stress adversely affects sleep duration** in the sampled population. Measures to reduce stress could improve sleep quality and overall health